

Marta Adamska

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PERSONAL PROFILE

Third year PhD student in sustainable AI systems with a Bachelor's degree in Computer Science. As a researcher I acquired a wide range of skills from creating LLM pipelines, programming and data analysis, to technical writing and presenting. I have experience working in commercial and academic projects which largely focus on sustainable development and reporting.

EDUCATION

PhD Computer Science October 2022 – Present
Lancaster University *Lancaster, UK*

- Currently in penultimate year of the programme
- Leading a project researching factors influencing Large Language Model energy usage during inference
- Building a system to measure energy consumption of Large Language Model's inference and performing an in-depth NLP analysis of the results
- Research showcases importance of prompt type and key phrases for energy consumption
- Currently researching possibilities of predicting energy consumption of LLMs
- Key technologies: Python, PyTorch, Transformers, Linux systems

BSc (Hons) Computer Science October 2019-July 2022
Lancaster University *Lancaster, UK*

- Grade: First Class (73%)
- Final Year Project: Art of Code - Exploring Approaches to Programming
- Relevant courses: Artificial Intelligence, Advanced Programming, Media Coding and Processing, Databases

PUBLICATIONS

M. Adamska, D. Smirnova, H. Nasiri, Z. Yu, P. Garraghan "Green Prompting" (under review) arxiv.org/abs/2503.10666
D. Smirnova, **M. Adamska**, Z. Yu, P. Garraghan "Energy-adaptive Model Selection in LLM-Orchestrated systems" (under preparation)

EXPERIENCE

Research Assistant December 2022 – April 2023
Lancaster University *Lancaster, UK*

- Working as a part of a research team to create a report on CO2 emissions of ICT systems across the UK
- Main tasks included researching relevant statistics and presenting the findings to the team
- Key skills: research, data analysis, presenting

Junior Test and Analysis Engineer April 2022 – October 2022
Sii Polska *Gdańsk, Poland*

- Analysing test results and creating reports on their success rate
- Improving and adding new features to report generation tool
- Key technologies: Python for data analysis, Linux systems

Research Assistant July 2021 – August 2021
Lancaster University *Lancaster, UK*

- Developing a dashboard displaying real-time resource consumption of a local data centre
- Key technologies: Docker, Grafana, Python

SKILLS

Programming Languages: Python, R, MATLAB, Java

Tools: LLMs, Git, Linux, SQL

Libraries: NumPy, SciPy, Pandas, PyTorch, TensorFlow, Transformers

Languages: Polish (Native), English (C1+), Mandarin Chinese (B2)

Soft skills: Problem-solving, time management, independent and team work, creativity and adaptability